The Sample Size Required in Importance Sampling

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March 9, 2026

Roadmap

1. Motivation and setup
2. Main theoretical message
3. Experimental study
4. Takeaways

Why this question matters

- We want expectations under a target law ν but can only sample from a proposal law μ.
- In high dimension or rare-event regimes, the likelihood ratio
$$\rho = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}$$
can be extremely uneven.
- Then a few samples dominate the estimator, while most samples contribute almost nothing.
- The practical question is simple but crucial:

How large must n be for importance sampling to be trustworthy?

Sample from proposal μ

↓

Compute weights $\rho(X_i)$

↓

Form weighted average

If weights are concentrated, one point can control the answer.

Importance sampling setup

Estimator

Let $X_1, \ldots, X_n \sim \mu$ i.i.d. and let

$$I(f) = \int f \, \mathrm{d}\nu, \qquad I_n(f) = \frac{1}{n} \sum_{i=1}^{n} f(X_i)\rho(X_i), \qquad \rho = \frac{\mathrm{d}\nu}{\mathrm{d}\mu}.$$

Then $I_n(f)$ is unbiased for $I(f)$.

Good regime : Many moderate weights, stable average, error decreases predictably.

Bad regime : Almost all weights tiny, a rare huge weight matters, estimator looks stable until it suddenly is not.

Core issue

The difficulty is driven by the mismatch between μ and ν, not just by a naive $1/n$ variance heuristic.

Classical variance logic can be misleading

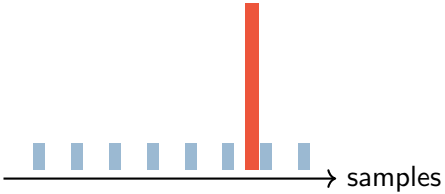
The common practical diagnostic is the empirical plug-in variance

$$v_n(f) = \frac{1}{n^2} \sum_{i=1}^{n} f(X_i)^2 \rho(X_i)^2 - \frac{I_n(f)^2}{n}.$$

People stop when $v_n(f)$ becomes small.

Problem: under μ, the rare large weights may not have appeared yet, so the empirical second moment can look tiny even when the estimate is still wrong.

Most observed weights are tiny



The key theorem: the exponential threshold

Let $Y \sim \nu$ and define

$$L = D_{\mathrm{KL}}(\nu \| \mu) = \mathbb{E}_\nu[\log \rho(Y)].$$

Theorem 1.1 (Chatterjee–Diaconis)

If $n = e^{L+t}$ with $t \geq 0$, then

$$\mathbb{E}\big|I_n(f) - I(f)\big| \leq \|f\|_{L^2(\nu)} \left(e^{-t/4} + 2\sqrt{\mathbb{P}(\log \rho(Y) > L + t/2)}\right).$$

Conversely, if $n = e^{L-t}$, then for every $\delta \in (0,1)$,

$$\mathbb{P}\big(I_n(1) \geq 1 - \delta\big) \leq e^{-t/2} + \frac{\mathbb{P}(\log \rho(Y) \leq L - t/2)}{1 - \delta}.$$

Interpretation

Up to concentration of $\log \rho(Y)$, the correct sample-size scale is $n \approx e^L$.

How to read the theorem

Sufficient side

If

$$\log n \gtrsim L,$$

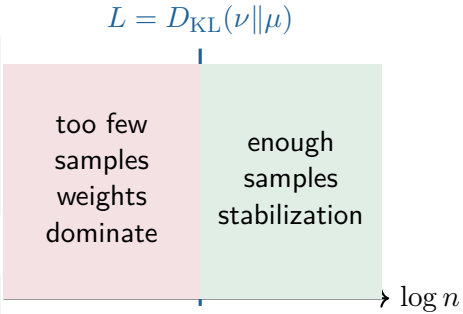
and $\log \rho(Y)$ concentrates around its mean, then the estimator is accurate.

Necessary side

If

$$\log n \lesssim L,$$

then even estimating $I(1) = 1$ can fail with non-negligible probability.



Insight: the sample complexity is information-theoretic. If μ and ν are far apart, the cost can be exponentially large.

Why Theorem 2.1 matters

Variance proxy can certify convergence far too early

For any $\varepsilon > 0$, there exists an integer

$$n \leq \varepsilon^{-2} 2^{1+\varepsilon^{-3}}$$

such that, for *every* pair (μ, ν) and every f with $\|f\|_{L^2(\nu)} \leq 1$,

$$\mathbb{P}\big(v_n(f) < \varepsilon\big) \geq 1 - 4\varepsilon.$$

- This bound depends mainly on ε.
- It does *not* track the real hardness of the pair (μ, ν).
- So the rule "stop when the empirical variance is small" can be almost universally over-optimistic.

Takeaway

A diagnostic is useful only if it reflects the same scale as the true convergence threshold. The empirical variance proxy does not.

A better practical diagnostic: weight concentration

Define

$$Q_n = \frac{\max_{1 \le i \le n} \rho(X_i)}{\sum_{i=1}^{n} \rho(X_i)}, \qquad q_n = \mathbb{E}(Q_n).$$

- Q_n measures how much a single weight dominates the normalized mass.
- If one sample carries a large fraction of the total weight, importance sampling is still degenerate.
- When the estimator is genuinely stable, Q_n should be small.

Theoretical support

If $\varepsilon_n = \mathbb{E}|I_n(1) - 1|$, then Theorem 2.3 implies that accurate estimation forces q_n to be small.

Limitations of the theory in practice

- The threshold uses
$$L = D_{\mathrm{KL}}(\nu \| \mu) = \mathbb{E}_\nu[\log \rho(Y)],$$
which is usually hard to compute because it depends on the target distribution.
- In Gibbs or exponential-family models, L often involves ratios of partition functions.
- Estimating L may require thermodynamic integration:
$$\log \frac{Z(\beta)}{Z(\beta_0)} = -\int_{\beta_0}^{\beta} \mathbb{E}_t[H] \, \mathrm{d}t,$$
which introduces discretization error, Monte Carlo error, and possible MCMC mixing issues.
- The theorem is about the correct *scale*; it is not designed as a sharp finite-sample confidence bound with tight constants.
- If $\log \rho(Y)$ does not concentrate well, the KL threshold becomes less predictive.

Experimental strategy

Setting	Goal	What is monitored?
Gaussian shift	ideal concentrated case	relative error, Q_n, $v_n(1)$, comparison with e^L
Rare-event mixture	multimodal / heavy-tail effects	whether KL still predicts success, and whether Q_n remains informative
Independent spins	Gibbs / partition function regime	phase transition around $\log n = L + c\sigma$, IID vs MCMC, true L vs estimated $\hat{L}$

Common question across all experiments

Which diagnostic aligns best with the real success or failure of importance sampling?

Experiment 1: Gaussian shift as the ideal case

Model

$$\mu = \mathcal{N}(m, I_d), \nu = \mathcal{N}(0, I_d), m = (a, \ldots, a)$$

Then

$$L = \frac{1}{2} d a^2.$$

Design

- $d = 50$
- $a \in \{0.3, 0.4, 0.5, 0.6\}$
- $n \in \{10^1, \ldots, 10^6\}$
- $R = 200$ replications

Expectation

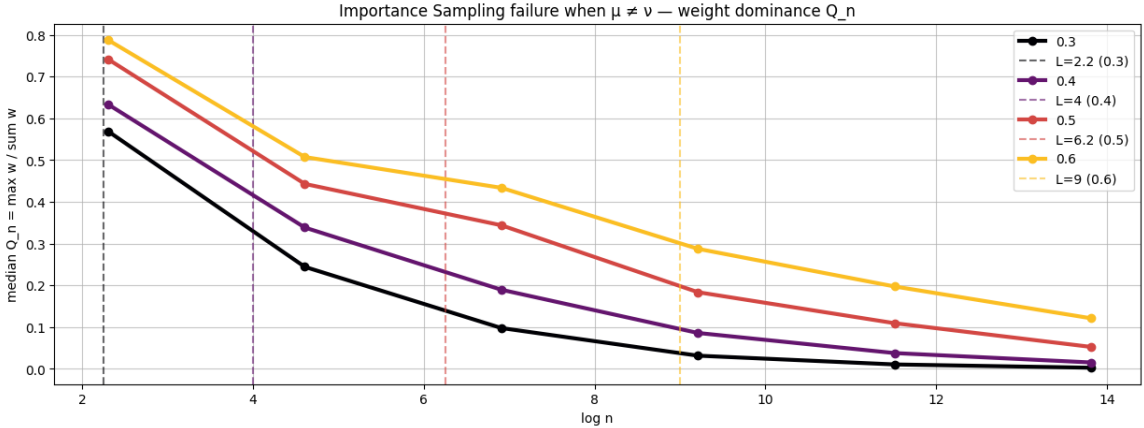
This is the best-case scenario for the theory:

- explicit KL divergence,
- concentration is good,
- the threshold $n \approx e^L$ should be visible.

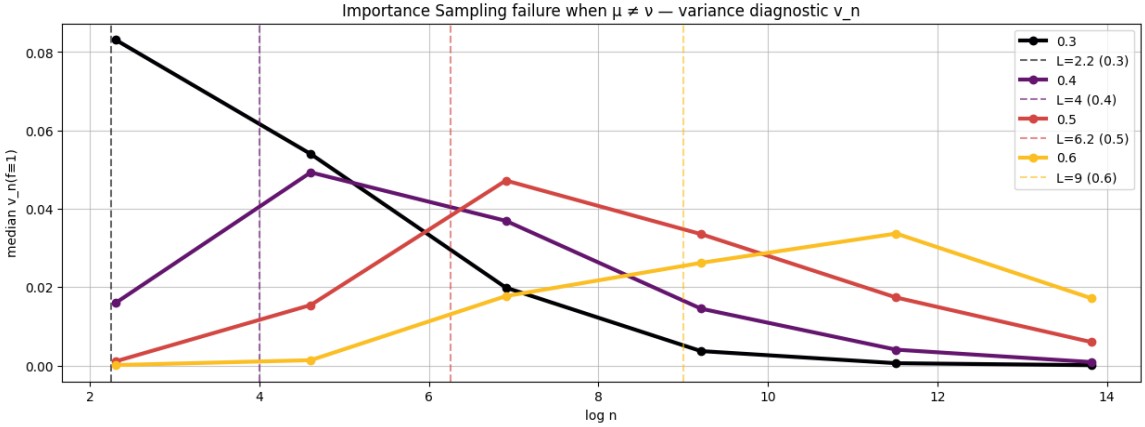
Question

Do the diagnostics separate real convergence from fake comfort?

Gaussian shift results



Gaussian shift results



Gaussian shift results interpretation

Observed pattern

- Relative error drops near the predicted threshold $L = \frac{1}{2}da^2$.
- Q_n decreases in the same region.
- $v_n(1)$ is much less informative and can look small too early.

Conclusion

In the concentrated Gaussian case, the KL threshold works very well and Q_n tracks it reliably.

Experiment 2: Rare-event mixture

Model

$$\nu = (1-\varepsilon)\,\mathcal{N}(-m, I_d) + \varepsilon\,\mathcal{N}(+m, I_d), \mu = \mathcal{N}(0, I_d)$$

- bimodality introduces a rare but important component,
- $L = D_{\mathrm{KL}}(\nu \| \mu)$ has no simple closed form,
- L is estimated by Monte Carlo:

$$L \approx \mathbb{E}_\nu \left[\log \frac{\mathrm{d}\nu}{\mathrm{d}\mu} \right].$$

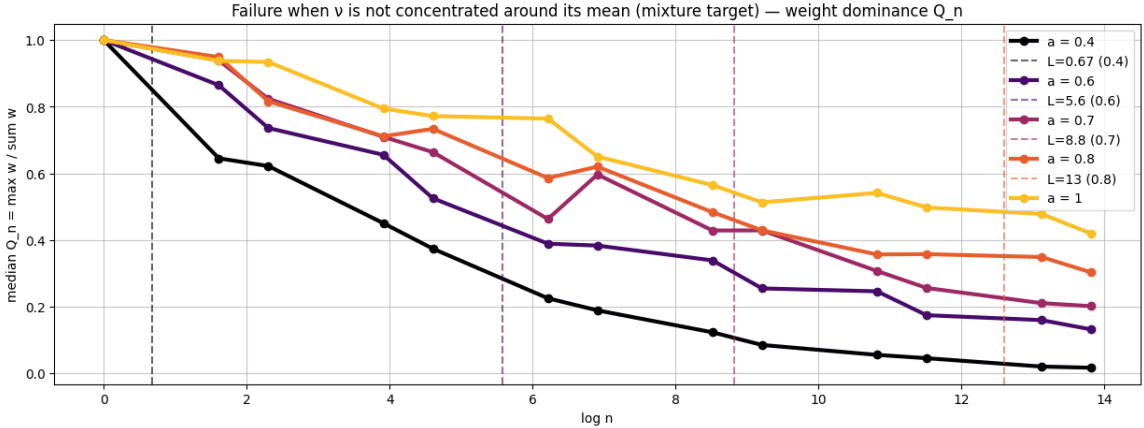
Why this case is harder

Even if the average KL gap is moderate, the target may not be concentrated around its mean behavior.

Expected lesson

The scalar quantity L may no longer be enough to predict performance sharply.

Rare-event mixture results



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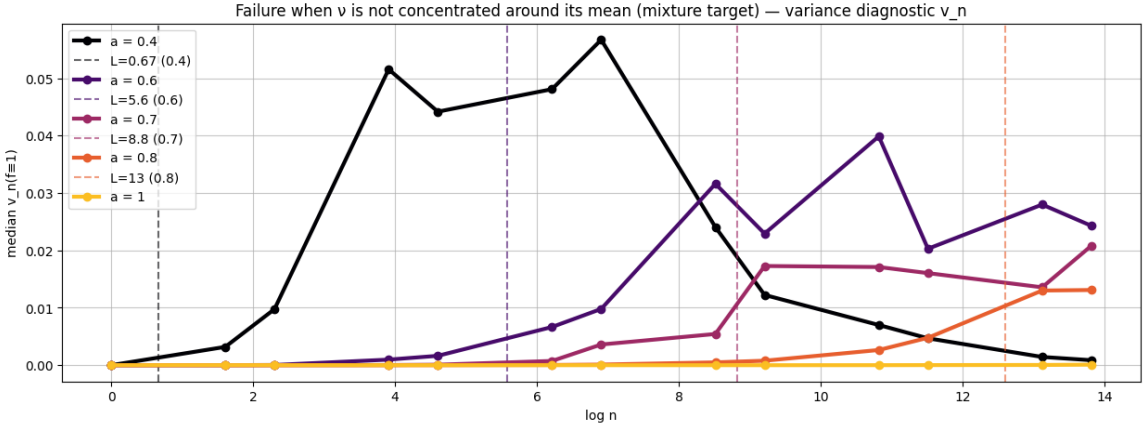


Rare-event mixture results interpretation

Observed pattern

- Large sample sizes relative to e^L can still fail.
- The relative error does not collapse as sharply as in the Gaussian shift case.
- Q_n remains a meaningful degeneracy signal.
- $v_n(1)$ again can be misleading.

Interpretation

When $\log \rho$ is not well concentrated, the KL threshold loses some predictive power.

Experiment 3: Independent spins / Gibbs setting

Model

$$H(x) = -\sum_i x_i, \qquad G_\beta(x) = \frac{e^{-\beta H(x)}}{Z(\beta)}.$$

The relevant quantities are

$$L = F(\beta_0) - F(\beta) - (\beta_0 - \beta)F'(\beta), \qquad \sigma = |\beta - \beta_0|\sqrt{F''(\beta)}.$$

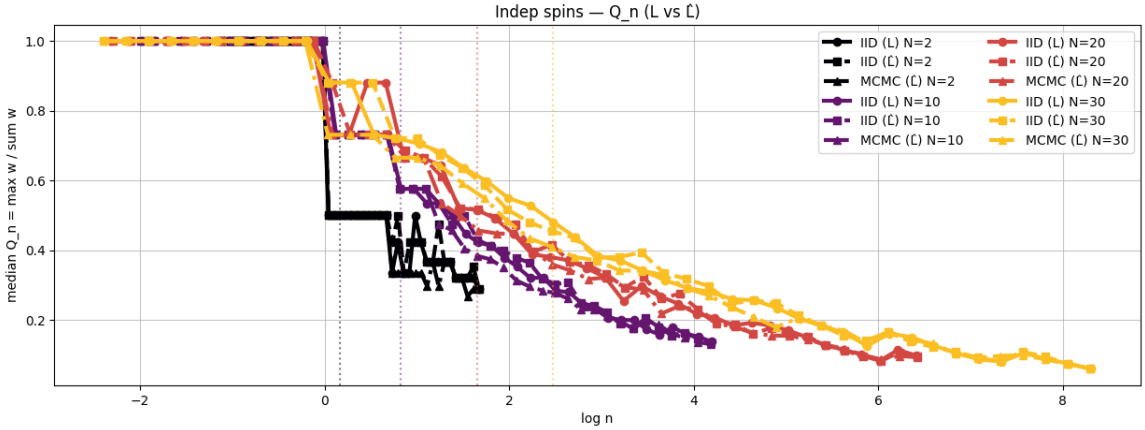
Protocol

- choose sample sizes through $\log n = L + c\sigma$,
- compare IID and Gibbs/MCMC sampling,
- compare true L with estimated $\hat{L}$ from thermodynamic integration.

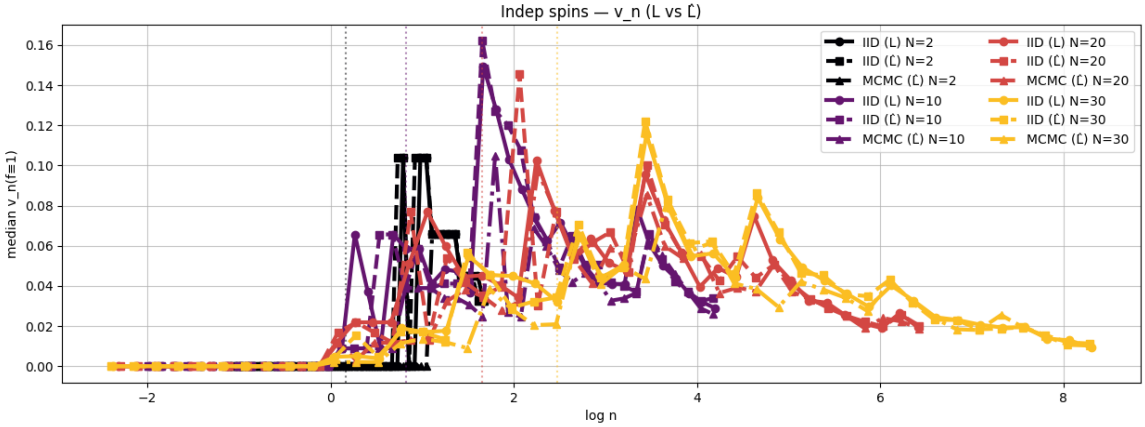
Why this matters

This is the practically relevant partition-function setting where L is not directly known.

Independent spins results



Independent spins results



Independent spins results interpretation

Observed pattern

- The phase transition remains centered around $\log n \approx L$.
- IID and MCMC give very similar qualitative behavior.
- Using $\hat{L}$ instead of L does not significantly move the threshold in this simple model.

Conclusion

The e^L scaling is robust in the independent-spin example.

Main takeaways

1. The right sample-size scale for importance sampling is typically

$$n \asymp e^{D_{\mathrm{KL}}(\nu \| \mu)}.$$

2. This is a much stronger and more informative principle than a naive variance-based rule.
3. The empirical variance proxy $v_n(1)$ can declare convergence far too early.
4. The weight-degeneracy statistic Q_n is much more faithful to the real success or failure of the method.
5. In concentrated models, the KL threshold is highly predictive; in rare-event or weakly concentrated targets, it becomes less sharp but still gives the right conceptual scale.
6. In Gibbs-style applications, estimating L is itself a nontrivial computational problem.

Practical recommendation

If you use importance sampling in practice

1. Do not trust a small empirical variance estimate on its own.
2. Inspect weight concentration — in particular, monitor Q_n or related effective-sample-size style diagnostics.
3. Whenever possible, reason about the KL mismatch between proposal and target.
4. If the problem is rare-event or multimodal, expect the true cost to be much higher than variance heuristics suggest.

Thank you

Appendix: theorem statements at a glance

Theorem 1.1:

- if $n = e^{L+t}$, the error is controlled by $e^{-t/4}$ plus the upper tail of $\log \rho(Y)$,
- if $n = e^{L-t}$, even the estimate of $I(1)$ may fail with substantial probability.

Theorem 2.1:

- for any tolerance ε, there exists a moderate sample size where $v_n(f) < \varepsilon$ with high probability,
- uniformly over all problem instances.

Theorem 2.3:

- if $I_n(1)$ is accurate, then $q_n = \mathbb{E}(Q_n)$ must be small.

Appendix: thermodynamic integration for L

In the Gibbs setting,

$$F(\beta) = \log Z(\beta), \qquad L = F(\beta_0) - F(\beta) - (\beta_0 - \beta)F'(\beta).$$

If F is not explicitly known, one may estimate it by

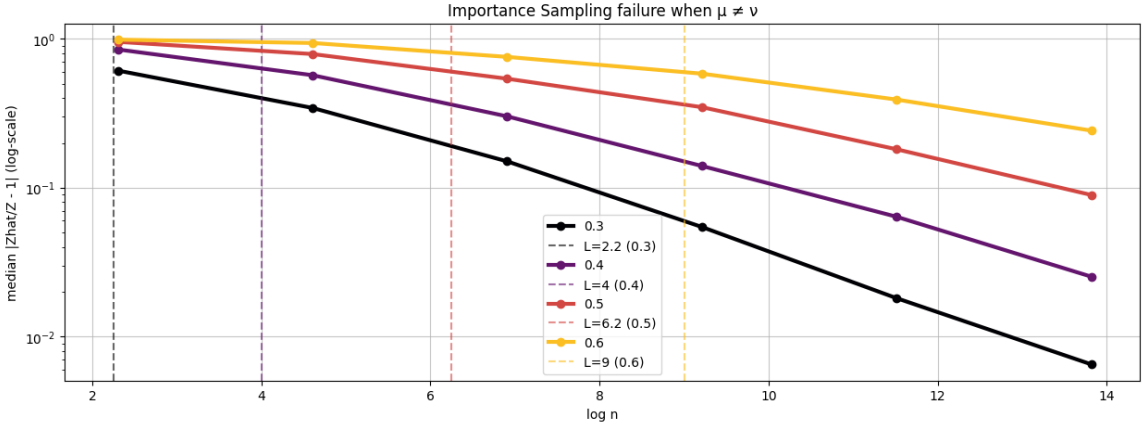
$$F(\beta) - F(\beta_0) = -\int_{\beta_0}^{\beta} \mathbb{E}_t[H] \, \mathrm{d}t.$$

- Simulate from intermediate laws G_t.
- Estimate $\mathbb{E}_t[H]$ over a grid of temperatures.
- Numerically integrate to get $\hat{L}$.

Caveat

This introduces a second layer of approximation before importance sampling even starts.

Gaussian shift results



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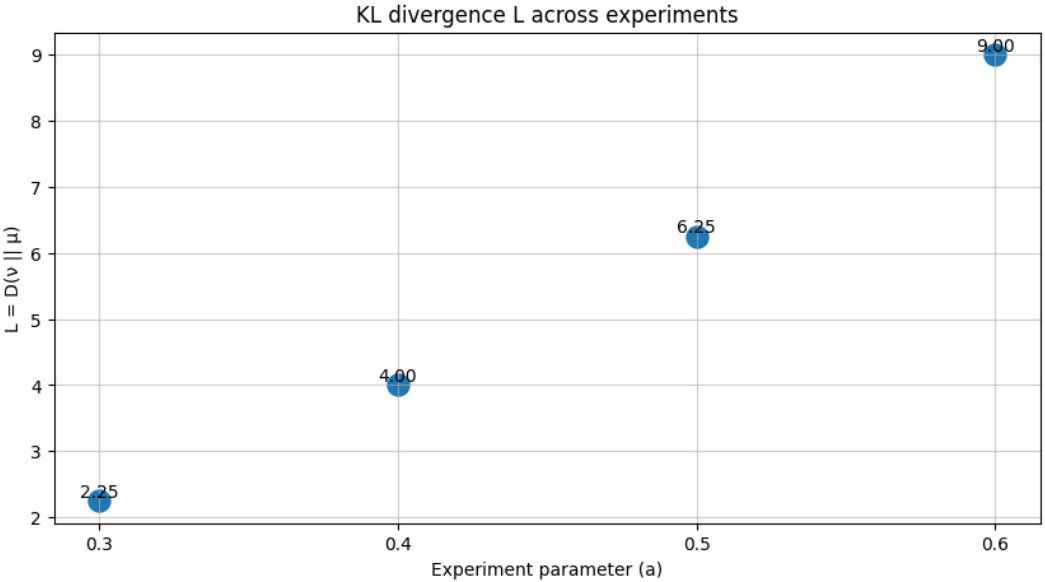
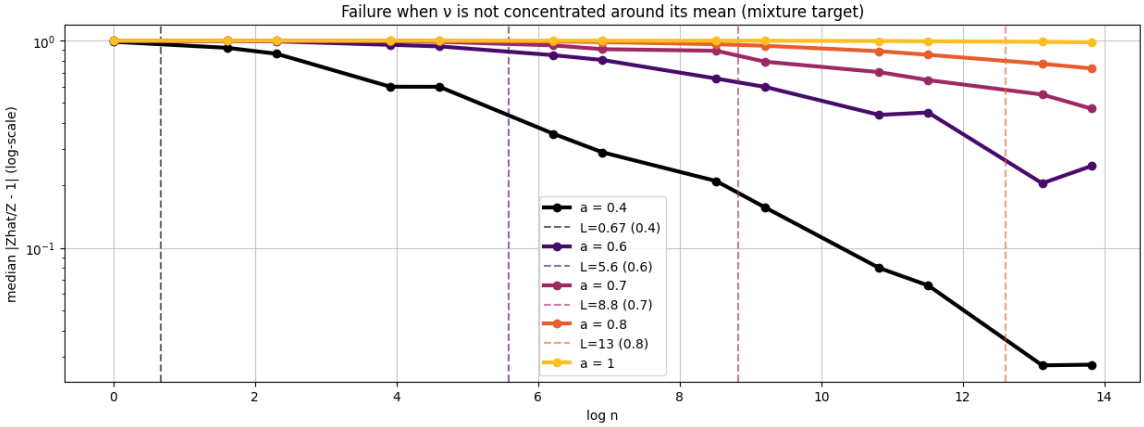
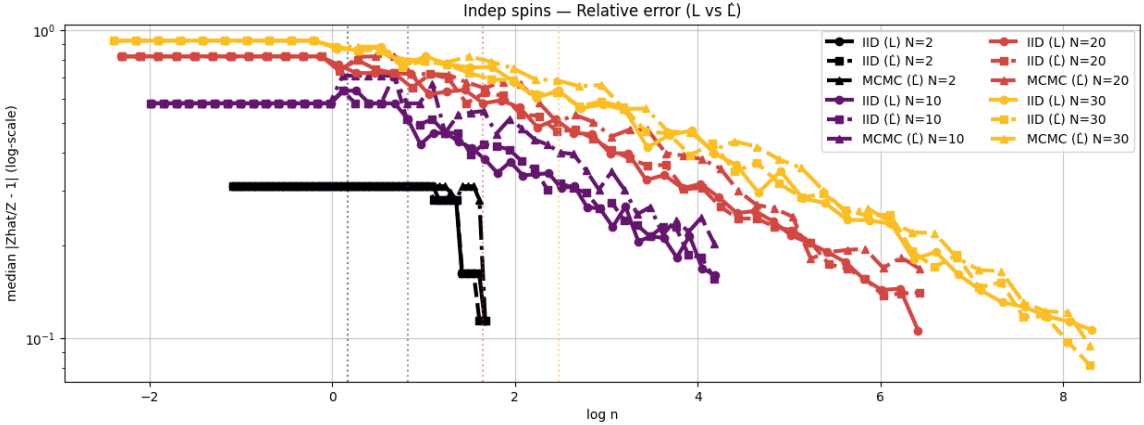


Rare-event mixture results



Independent spins results



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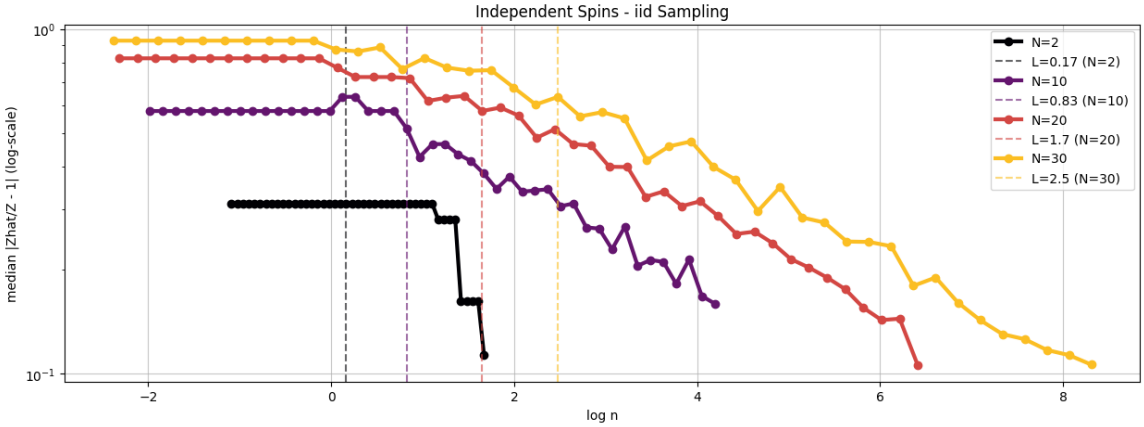

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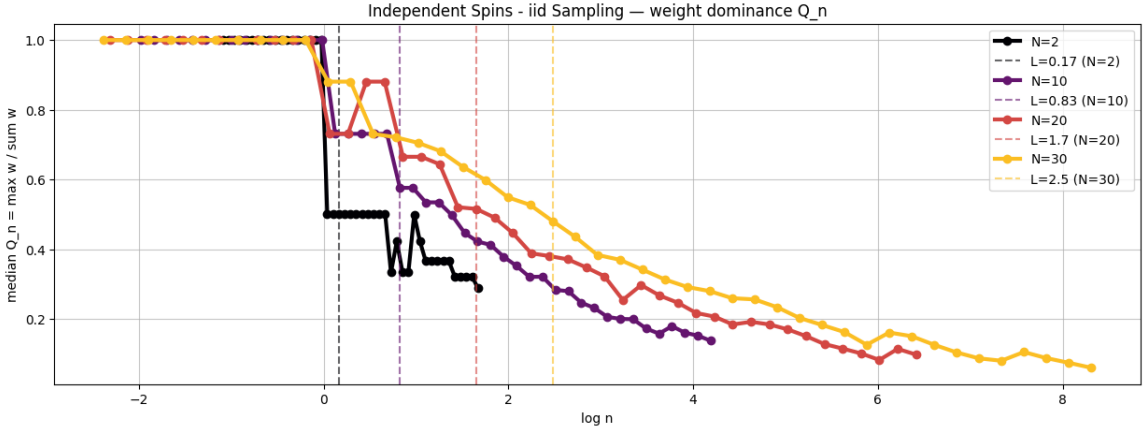
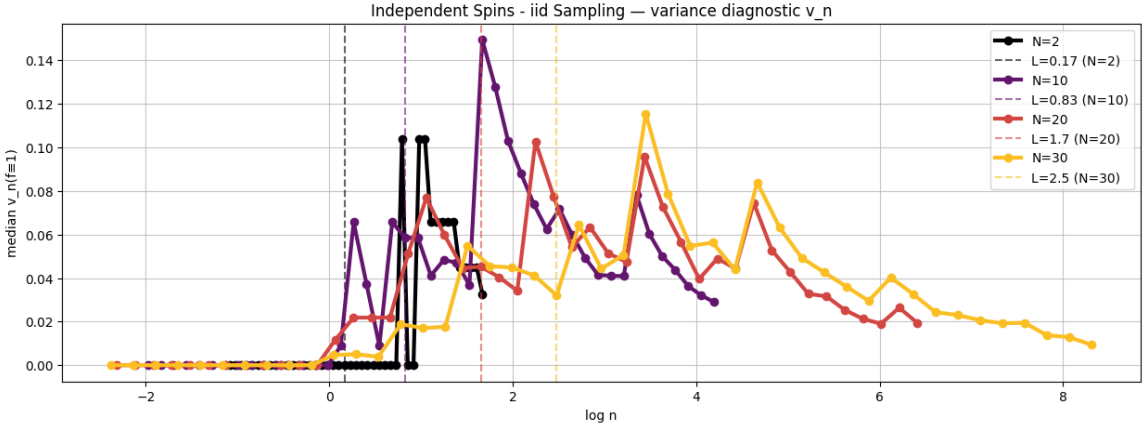
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